Redox mechanisms: from ion motion to memory

An oxide memory is a mixed ionic–electronic system. Its electrical response depends on mobile defects, electrode reactions, thermal history and measurement time. This unit develops the conditions needed to connect an observed I–V curve to a physical mechanism.

All numerical examples are hypothetical teaching models; they are not device specifications or reported measurements.

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Separate three observations

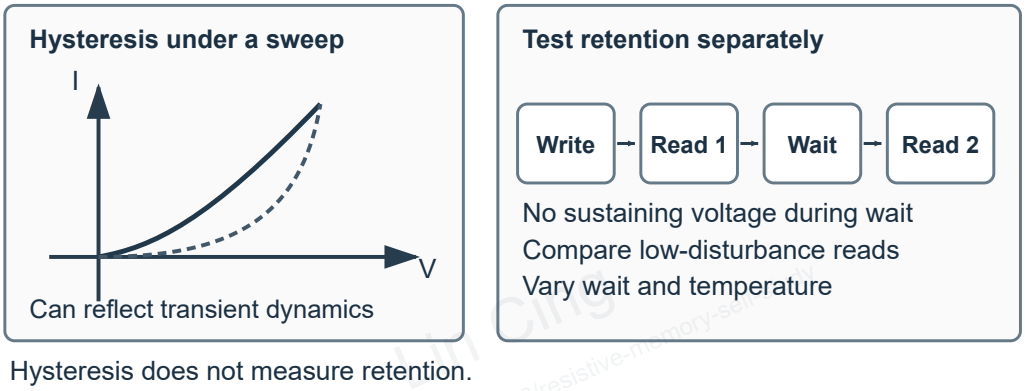
Switching is a change in conductance caused by an operation. **Retention** asks whether a distinguishable state remains after the driving signal is removed. **Short-term hysteresis** means the response depends on recent history over the measurement time.

A loop in a voltage sweep can occur when ion and electron responses have different timescales. It does not, by itself, establish long-term nonvolatile memory. Conversely, two retained states can be distinguished using a suitably non-disturbing read protocol without relying on the shape of a continuous sweep.

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A measurement distinguishes timescales

I–V hysteresis alone does not prove nonvolatile memory



Hysteresis does not measure retention.

The sketch compares two protocols; it does not assert a specific material or quantitative retention time. The waiting period must be specified and the read pulse checked for disturbance.

From material change to terminal current

One possible chain is: an applied pulse changes an electrode reaction and/or moves ionic defects; a nonuniform composition develops; that distribution modifies an electronic conduction path or interface barrier; the read current changes.

These links are hypotheses to be tested. An I–V loop alone cannot tell whether conduction is filamentary, interface-controlled, or distributed across the film. A mechanism should also predict spatial, chemical and time-dependent observations.

Charge carriers and their roles

The electronic current can be dominated by electrons or holes, while much slower ionic defects alter the device structure. The measured terminal current can include electronic, ionic and displacement contributions:

$$I_{\text{total}} = I_e + I_{\text{ion}} + I_{\text{disp}}.$$

In a one-dimensional series cross section, charge continuity gives the same total current at each position; individual contributions can vary with position and time as charge accumulates. An ammeter does not automatically measure ion flow.

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Reaction bookkeeping: an oxygen vacancy

For oxygen leaving an oxide into an oxygen reservoir, one possible idealized reaction is

$$O_O^{\times} \rightleftharpoons \tfrac{1}{2}O_2 + V_O^{\bullet\bullet} + 2e'.$$

Here $O_O^{\times}$ is oxygen on its lattice site, $V_O^{\bullet\bullet}$ is a vacancy with effective charge +2 relative to the lattice, and e' denotes an electron with effective charge −1. Both atom count and effective charge balance. This is a model of one charge state, not a claim that all vacancies have this charge.

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When may vacancy concentration predict electrons?

If vacancies are the only significant positively charged defects, electrons the only significant negative carriers, and local charge neutrality applies, then $n \simeq 2[V_O^{\bullet\bullet}]$.

For $[V_O^{\bullet\bullet}] = 2 \times 10^{18}\,\mathrm{cm}^{-3}$, the model gives $n \simeq 4 \times 10^{18}\,\mathrm{cm}^{-3}$. With negatively charged acceptors included, neutrality becomes $2[V_O^{\bullet\bullet}] = n + N_A^-$: for $[V_O^{\bullet\bullet}] = 3 \times 10^{18}$ and $N_A^- = 2 \times 10^{18}\,\mathrm{cm}^{-3}$, $n = 4 \times 10^{18}\,\mathrm{cm}^{-3}$ instead of $6 \times 10^{18}\,\mathrm{cm}^{-3}$ when acceptors are incorrectly omitted. Interface space charge can also invalidate local neutrality.

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ECM: metal transfer and reduction

In an electrochemical metallization (ECM) cell with an active Ag electrode, a possible oxidation reaction is $\mathrm{Ag} \rightarrow \mathrm{Ag}^+ + e^-$. Ag ions enter the switching layer; reduction $\mathrm{Ag}^+ + e^- \rightarrow \mathrm{Ag}$ can deposit metal elsewhere.

Where growth begins and how a path forms depend on ion transport, electron supply, interfaces and rates. The equations do not imply that every ECM cell has the same filament-growth direction or that SET has one universal polarity.

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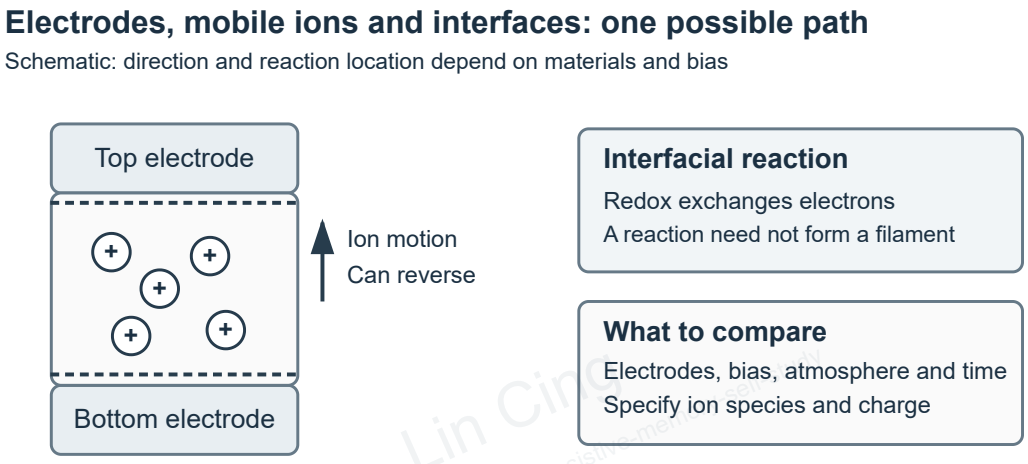
VCM: oxygen exchange and vacancy redistribution

In a valence-change mechanism (VCM), oxygen chemical potential, vacancy populations and local transition-metal valence can change. A reduced path, a narrow gap near an electrode, or a modified interface region can alter electronic transport.

VCM need not mean a metallic filament; ECM need not make every observed filament metallic throughout its lifetime. Composition-sensitive and spatially resolved measurements are needed to distinguish models.

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Follow ions and interfaces in a cell



The illustration shows a possible direction of ion motion, not an assigned electrode polarity or a specific reaction. For an Ag-containing ECM stack, the reactions on the preceding page are examples; actual growth direction depends on the stack, voltage and kinetics.

Electrochemical potential and driving force

For species i with charge number z_i, electric potential ϕ and chemical potential μ_i, the electrochemical potential is

$$\widetilde{\mu}_i = \mu_i + z_i e \phi.$$

At fixed temperature, a gradient of $\widetilde{\mu}_i$ drives transport. A concentration gradient contributes through μ_i; the electric field $E = -\partial\phi/\partial x$ contributes through $z_i e\phi$. Electric field alone does not determine net transport when concentration gradients are present.

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Drift and diffusion in a limited model

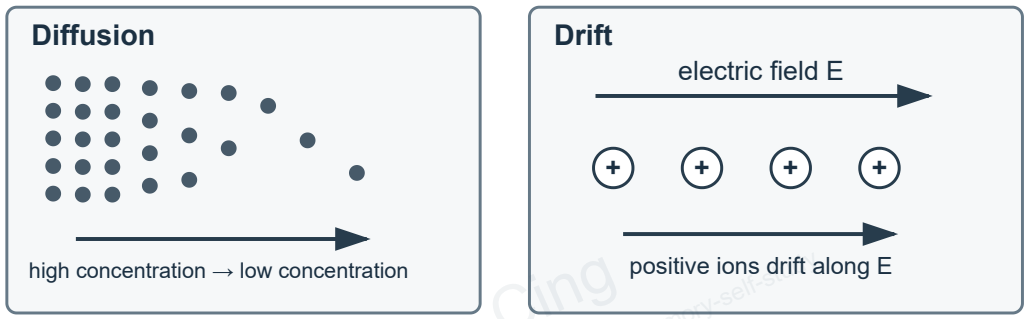
For dilute, noninteracting particles with concentration $c(x)$, constant D, charge number z, and the Einstein relation, the one-dimensional number flux is

$$J = -D\frac{\partial c}{\partial x} + \frac{Dze}{k_B T} cE.$$

Here J is number per area per time, not electrical current density; the latter is $j_{\mathrm{ion}} = zeJ$. For concentrated defects, strong fields, interactions or nonuniform temperature, use an appropriate nonideal transport model rather than this expression unchanged.

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Compare diffusion with electric drift



Net flux combines both terms and boundary conditions; electron flow is separate.

The arrows are for the stated positive defect and illustrated gradients only. Changing charge sign or either gradient can reverse an individual contribution; the net flux is their vector sum.

Drift and diffusion: calculate both terms

For the dilute model, let $D = 10^{-18}\,\mathrm{m^2/s}$, $z = 2$, $E = 10^{8}\,\mathrm{V/m}$, $c = 10^{24}\,\mathrm{m^{-3}}$, $dc/dx = 10^{30}\,\mathrm{m^{-4}}$ and $k_B T = 4.14 \times 10^{-21}$ J.

The diffusion flux is $-10^{12}\,\mathrm{m^{-2}s^{-1}}$. With $e = 1.60 \times 10^{-19}$ C the drift contribution is $+7.73 \times 10^{15}\,\mathrm{m^{-2}s^{-1}}$; at this precision the net flux is also approximately $+7.73 \times 10^{15}\,\mathrm{m^{-2}s^{-1}}$. These are particle fluxes, not amperes.

A field does not erase a migration barrier

Suppose a 12 nm film has a uniform 0.6 V drop. The electric-field magnitude is 0.05 V/nm. Across a 0.2 nm hop along the electric field the potential falls by 0.01 V; for a hypothetical $z = +2$ defect the endpoint electrostatic energy changes by -0.02 eV.

This compares endpoint energies. The reduction of the actual activation barrier depends on the transition-state position and energy landscape and is not automatically 0.02 eV. A highly nonuniform drop at an interface changes the estimate.

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Interface reaction versus bulk transport

An electrode can exchange oxygen with the oxide, store oxygen, react to form an interlayer, or approximately block exchange. A blocking boundary constrains normal flux there; it does not imply that the defect concentration is uniform inside the film.

If supply at one interface is slow compared with redistribution in the oxide, defects can accumulate in a different region from the fast-supply case. Growth direction and the active gap therefore depend on **both** boundary reaction kinetics and bulk transport, not electrode name alone.

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A supply–transport balance, with limits

For one illustrative slab of thickness L and cross-sectional area A, let $N(t)$ be its spatially averaged defect concentration, with injection flux s and removal flux r on opposite boundaries. With no net volumetric reaction, conservation gives $d(ANL)/dt = A(s - r)$, hence

$$\frac{dN}{dt} = \frac{s - r}{L}.$$

Fluxes may depend on concentration, voltage, temperature and time. This averaged balance cannot determine a local profile, filament geometry, or which side the filament grows from; those require the transport equation and boundary conditions.

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Crossing an assumed concentration threshold

Take the simplified constant-flux model with $L = 10\ \text{nm} = 10^{-6}$ cm, $s = 8 \times 10^{12}$ and $r = 2 \times 10^{12}\ \text{cm}^{-2}\text{s}^{-1}$. Then $dN/dt = 6 \times 10^{18}\ \text{cm}^{-3}\text{s}^{-1}$.

If $N(0) = 2 \times 10^{18}\ \text{cm}^{-3}$ and a specified model threshold is $N_c = 5 \times 10^{18}\ \text{cm}^{-3}$, then $N(t) = N(0) + (s - r)t/L$ reaches N_c after $(5 - 2)/6 = 0.5$ s. This is an average crossing, not proof of a local metal transition; a real threshold depends on material and spatial distribution.

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A separate steady-state spatial model

Now consider a different one-dimensional model: constant D, no drift, no volume sources, and fixed endpoint concentrations. Steady conservation gives $dJ/dx = 0$, and Fick's law gives $J = -D\,dc/dx$. Thus $c(x)$ is linear and the flux is uniform.

Take $L = 10\ \text{nm} = 10^{-6}$ cm, $D = 10^{-14}\,\text{cm}^2/\text{s}$, $c(0) = 8 \times 10^{18}$ and $c(L) = 2 \times 10^{18}\,\text{cm}^{-3}$. Then

$$c(x) = \left(8 - 6\frac{x}{L}\right) 10^{18}\,\text{cm}^{-3}, \quad J = +6 \times 10^{10}\,\text{cm}^{-2}\text{s}^{-1}.$$

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Where does the local threshold hold?

Using that steady diffusion solution and the illustrative threshold $N_c = 5 \times 10^{18}\,\text{cm}^{-3}$, solve $8 - 6x/L \geq 5$: the interval is $0 \leq x \leq L/2$, or 0–5 nm, of length 5 nm.

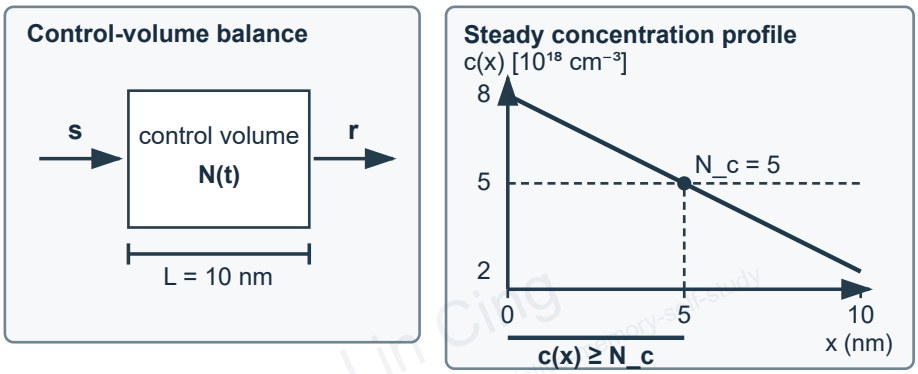
The assumed boundary concentrations and steady-state profile differ from the earlier transient control-volume example; do not combine the two as one time history. Crossing a chosen concentration threshold in this 1D model is not a general prediction of a connected, metallic filament.

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Boundary flux and local profile

Control-volume balance and a concentration profile

Teaching examples: the two models are separate



A threshold alone proves neither phase change nor filament.

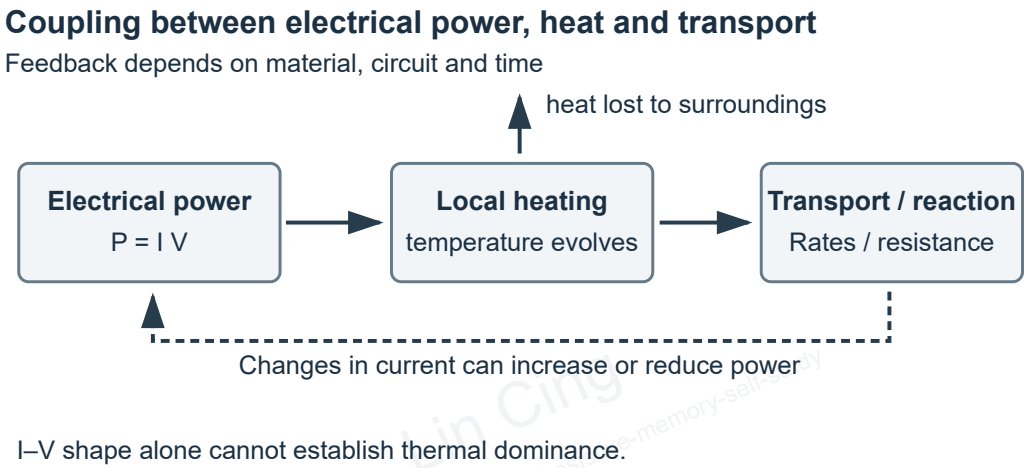
The left panel is a transient average; the right panel is a separate fixed-boundary steady state. Their fluxes and concentrations are not one continuous experimental sequence.

Forming, compliance and feedback

Some devices require electroforming: an initial electrical treatment that establishes a switchable state. Field-driven reactions and local Joule heating can reinforce one another; the evolving structure changes the local field and current.

Compliance limits the terminal current under the stated circuit and timing. Finite response time and parasitic capacitance can still produce a transient overshoot, so the set point alone does not fix local peak current, temperature or filament dimensions. Other devices are forming-free under their specified preparation and test conditions.

Feedback can amplify a local change



I–V shape alone cannot establish thermal dominance.

This loop is a possible mechanism under a specified circuit: increasing temperature need not increase conductance in every material or bias regime. Compliance and thermal loss limit the process differently.

Why a nanometre gap matters

Take a deliberately simple model in which read current through a barrier is $I(g) = I_0 e^{-2\kappa g}$ with fixed I_0 and $\kappa = 5\,\mathrm{nm}^{-1}$. A gap change from $g_1 = 1.0$ nm to $g_2 = 1.3$ nm gives

$$\frac{I(g_2)}{I(g_1)} = e^{-2(5)(0.3)} = e^{-3} \simeq 0.0498.$$

Thus the wider-gap current is about 5% of the initial current. This is an exponential sensitivity illustration, not a complete tunnelling theory: barrier height, bias, geometry and prefactor may also change.

Interface control is not proof of a filament

An interface barrier can limit the current even when the remainder of the oxide is relatively conductive. Changes in charge distribution and local doping can modify barrier width, band bending and available transport channels.

At fixed temperature and device geometry, ask whether an area series, local current map, spectroscopy and cross-sectional imaging support a distributed interface or localized path. No single area-scaling result proves the entire current topology without checking confounders.

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Bipolar and unipolar are operating descriptions

In **bipolar** operation, the two transitions are induced under opposite voltage polarities as defined for that device. In **unipolar** operation, both can occur at the same polarity under different pulse, current, thermal or circuit conditions.

The label classifies a protocol, not a unique microscopic mechanism. In particular, same-polarity RESET does not prove a particular temperature or diffusion process without supporting measurement.

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Compare the energy of voltage scans

For a voltage ramp beginning at zero with constant positive scan rate $v = dV/dt$, substitute $dt = dV/v$ into $W = \int V I \, dt$ to obtain

$$W = \frac{1}{v} \int_0^{V_{\mathrm{end}}} V I(V) \, dV.$$

Use the current actually measured along each scan, including its state history; this change of variables does not assume SET and RESET share the same I–V curve. Different ramp rates and switching points must be included explicitly.

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A same-polarity scan comparison

Take two hypothetical positive-going scans, each at $v = 1$ V/s. For the SET scan let $I = aV$ with $a = 1\,\mu$S up to 1 V; for RESET use $a = 4\,\mu$S up to 0.5 V. Integrating $VI = aV^2$ gives

$$W_{\mathrm{SET}} = \frac{1\,\mu\mathrm{S}}{1\,\mathrm{V/s}} \frac{1^3}{3} = \tfrac{1}{3}\,\mu\mathrm{J}, \quad W_{\mathrm{RESET}} = \frac{4\,\mu\mathrm{S}}{1\,\mathrm{V/s}} \frac{0.5^3}{3} = \tfrac{1}{6}\,\mu\mathrm{J}.$$

Higher current at a given voltage alone does not determine integrated energy when voltage ranges and durations differ. Even if one scan delivers more energy, that alone cannot determine which reaches a higher peak temperature: the heat-loss path and time dependence also matter.

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Power, energy and temperature are different

Instantaneous electrical power is $P(t) = V(t)I(t)$ and delivered energy is $W = \int P(t)dt$. The same energy can produce different peak temperatures if pulse widths, heat loss and thermal paths differ.

For a lumped model, $C_{\mathrm{th}}\, d\Delta T/dt = P(t) - \Delta T/R_{\mathrm{th}}$. With constant power, $\Delta T(t) = PR_{\mathrm{th}}(1 - e^{-t/(R_{\mathrm{th}}C_{\mathrm{th}})})$. This model is not a spatial temperature map.

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A thermal pulse worked through

Assume $V = 1$ V and $I = 100\,\mu$A remain constant, so $P = 0.1$ mW. Let $R_{\mathrm{th}} = 10^6$ K/W, $\tau_{\mathrm{th}} = 100$ ns, and initially $\Delta T = 0$. The implied $C_{\mathrm{th}} = \tau_{\mathrm{th}}/R_{\mathrm{th}} = 10^{-13}$ J/K; the steady rise is $PR_{\mathrm{th}} = 100$ K.

After 100 ns the idealized rise is $100(1 - e^{-1}) \simeq 63.2$ K. Changing the thermal resistance, heat capacity or pulse shape changes the result; measuring electrical energy alone does not reveal peak local temperature.

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Retention and activation: a toy estimate

An activated process is often approximated by $\tau = \tau_0 \exp(E_a / k_B T)$ when one rate-limiting barrier dominates and temperature is constant. For $\tau_0 = 10^{-12}$ s, $E_a = 1.0$ eV and $k_B T = 0.02585$ eV at 300 K,

$$\tau \simeq 10^{-12} e^{38.68} \simeq 6.3 \times 10^{4}\ \mathrm{s} \approx 17.5\ \mathrm{h}.$$

This hypothetical time is not a retention prediction for a particular device. Real retention depends on barrier distributions, geometry, read criterion and other mechanisms; the fitted prefactor must be stated.

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Read–wait–read: test memory directly

Program two populations to HRS and LRS, remove the programming bias, wait at controlled temperature, then apply the same low-disturb read to each. Repeat for multiple times and devices; report read voltage, pulse duration, temperature and the state distributions.

Retention is a criterion on the separation and error rate of those distributions, not merely a single resistance value. Check that repeated reads do not themselves change the state; include drift and uncertainty.

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What an I–V crossing can and cannot show

In a simplified passive, non-battery-like device model, negligible displacement and ionic currents at low frequency can make the terminal curve pass near the origin. Even if a curve passes exactly through the origin, that is **not sufficient** to establish retention; an internal state may relax.

Offsets from capacitance, electrochemical electromotive forces, instrumentation and finite sweep rate can shift the crossing. These assumptions must be checked before interpreting a crossing. Compare sweep rates and perform the read–wait–read experiment.

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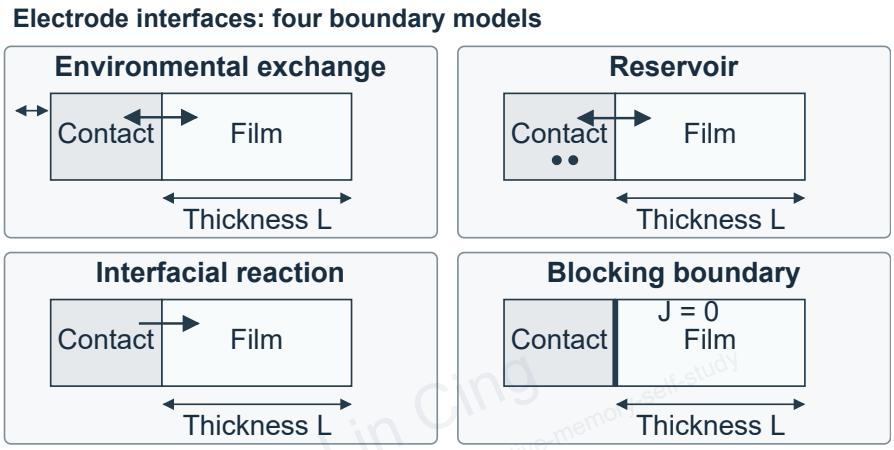
Build a falsifiable mechanism

1. Specify electrode stack, polarity convention, forming and compliance circuit, pulse timing, atmosphere and temperature.
2. Compare hypotheses using predictions for spatial current, element distribution, state evolution and temperature—not only a fit to one I–V curve.
3. Test alternative explanations and controls, including interface changes, parasitic current, measurement history and sample-to-sample variation.

An original model is useful when it explains the data within its assumptions and can be challenged by another experiment.

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Electrode boundaries determine the oxygen inventory



Boundary depends on material and bias; arrows show possible flux.

An electrode may allow exchange with the environment, store oxygen in grain boundaries, react to form a reservoir layer, or approximately block exchange. The substrate, edge leakage and actual interface matter: a nominally inert electrode is not necessarily perfectly blocking. These labels describe a particular stack and timescale, not permanent material constants.

Count oxygen and vacancies at the boundaries

For an oxide slab of thickness L, take vacancy number flux $J(0)$ as entering at $x = 0$ and $J(L)$ as leaving at $x = L$. With no volume generation or reaction, integrating $\partial c/\partial t = -\partial J/\partial x$ gives

$$\frac{d\bar{c}}{dt} = \frac{J(0) - J(L)}{L}.$$

In a closed, blocking stack, both boundary fluxes vanish and the total vacancy inventory is conserved, although its *spatial distribution* can change. An interfacial reservoir or a reaction adds its own inventory and kinetics; include these in the total balance rather than assuming a fixed oxide inventory. Oxygen and vacancy fluxes require a consistent sign convention and are not interchangeable without a defect model.

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Quantify net boundary exchange

For example, a net inward vacancy flux $J(0) - J(L) = 10^{12}\,\mathrm{cm^{-2}s^{-1}}$ across $L = 10$ nm raises the average concentration by $2 \times 10^{18}\,\mathrm{cm^{-3}}$ in 2 s. A possible linear exchange boundary at $x = 0$ is $J(0) = k[c_{\mathrm{eq}} - c(0)]$, with k in cm/s; its parameters and the opposite boundary must be specified to solve the spatial profile.

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Supply versus transport: compare two limiting pictures

Take the book's particular forming convention: cathode at $x = 0$ on an oxygen-blocking substrate, anode at $x = L$. If vacancy supply at the anode is limited while bulk removal is relatively fast, a high-vacancy region can accumulate toward the blocking cathode and its boundary may move toward the anode. If anode supply remains relatively fast compared with bulk removal, a high-concentration region can instead extend from the anode while a low-concentration gap remains near the cathode.

These are conditional qualitative pictures, not a formula that maps polarity to a unique front. Rate relationships may reverse as conductivity and fields change, and the assumed threshold may never be reached. Electrode geometry, reactions and the actual field profile must be measured or modelled.

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Forming and RESET are different boundary problems

In one bulk-reduction scenario, forming increases vacancies without requiring the whole oxide to cross the assumed metallic threshold. During a subsequent RESET, the former anode can become the cathode; fast vacancy removal at that boundary can leave a local low-concentration, high-resistance gap. A later SET may replenish it under the stated reactions and boundary conditions.

This is one proposed explanation, not a prediction for every device. Reversing the terminal voltage changes electrode roles only under the defined voltage convention; a Schottky barrier, interfacial reactions, heat or another transport pathway can instead dominate the observed switching.

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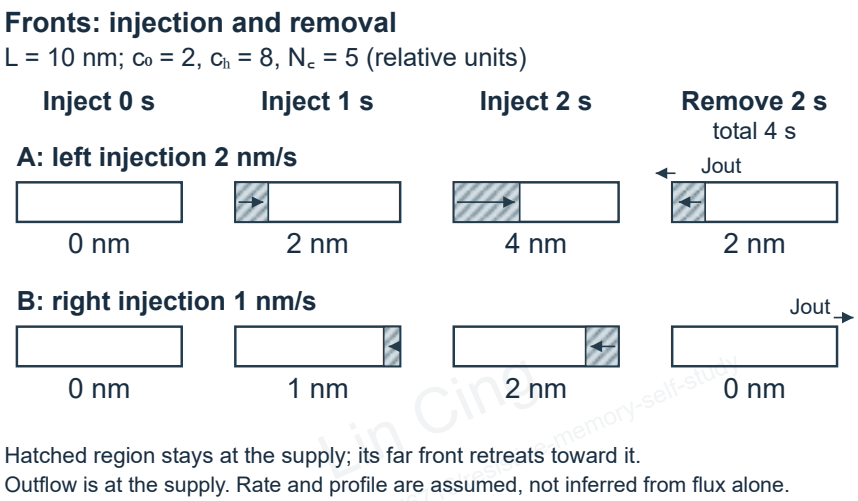
A conservation-based moving-front example

For a separate illustrative sharp-front model, let $L = 10$ nm, $c_{\mathrm{low}} = 2$ and $c_{\mathrm{high}} = 8$ in units of $10^{18}\,\mathrm{cm}^{-3}$, and let the chosen threshold be $c_{*} = 5$ in the same units. In case A, the high region starts at $x = 0$ and grows at 2 nm/s; in case B it starts at $x = L$ and grows at 1 nm/s. At $t = 2$ s, the above-threshold lengths are 4 and 2 nm, respectively.

Before the front reaches a boundary, conservation requires net influx per area $s = (c_{\mathrm{high}} - c_{\mathrm{low}})v$: 1.2×10^{12} and $6 \times 10^{11}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}$. The model prescribes both the front and its speed; it does not derive the moving front from the book's microscopic kinetics.

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Follow the two prescribed fronts



Hatched region stays at the supply; its far front retreats toward it.
Outflow is at the supply. Rate and profile are assumed, not inferred from flux alone.

The two configurations have separate boundary conditions. Arrows and front speeds are assumptions of this example, not a universal relationship between growth direction and terminal polarity.

Track the RESET front without inventing a mechanism

Now assume each high-concentration region remains anchored at its original supply interface, and its far front recedes toward that interface at 1 nm/s for 2 s. Assume internal redistribution maintains this shape and the corresponding net loss leaves through that interface. The lengths change A: $4 \to 2$ nm and B: $2 \to 0$ nm; A retains $[0, 2]$ nm. The magnitude of the net outward vacancy flux is $(8 - 2) \times 10^{18} \times 10^{-7} = 6 \times 10^{11}\,\mathrm{cm^{-2}s^{-1}}$ while the front exists.

This geometry and redistribution are deliberately assigned; the net flux alone does not uniquely determine either. This toy history is separate from the qualitative source-limited and transport-limited pictures above. An above-threshold interval in one dimension alone does not prove a three-dimensional connected filament, a metallic phase, or a unique SET/RESET polarity.

References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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